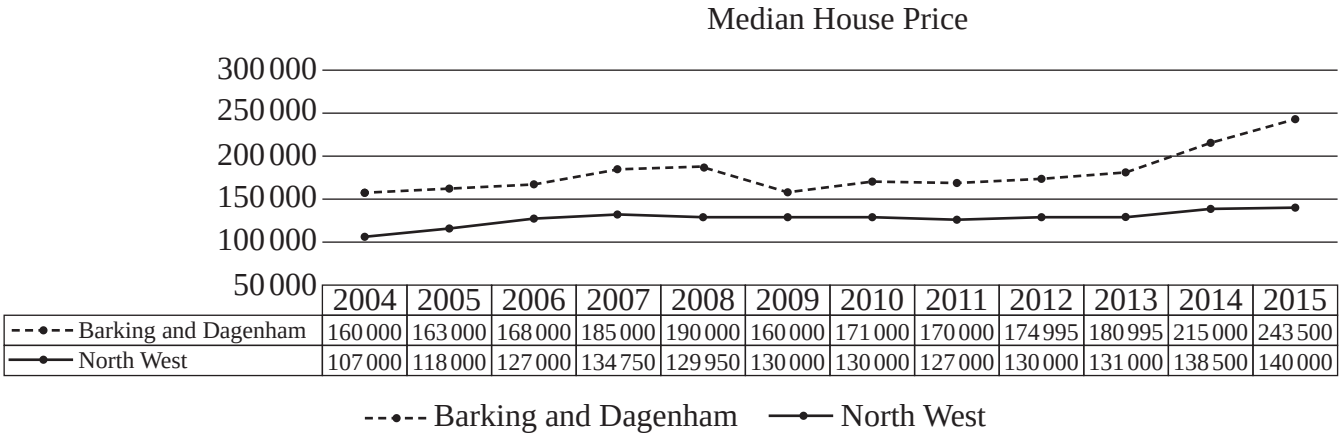
11 The pre-release material contains information concerning median house prices over the period 2004 – 2015. A spreadsheet has been used to generate a time series graph for two areas: the London borough of "Barking and Dagenham" and "North West". This is shown together with the raw data in Fig. 11.1.



	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015
Barking and Dagenham	160 000	163 000	168 000	185 000	190 000	160 000	171 000	170 000	174 995	180 995	215 000	243 500
North West	107 000	118 000	127 000	134 750	129 950	130 000	130 000	127 000	130 000	131 000	138 500	140 000

Fig. 11.1

Dr Procter suggests that it is unusual for median house prices in a London borough to be consistently higher than those in other parts of the country.

(a) Use your knowledge of the large data set to comment on Dr Procter's suggestion. **[1]**

Dr Procter wishes to predict the median house price in Barking and Dagenham in 2016. She uses the spreadsheet function LINEST to find the equation of the line of best fit for the given data. She obtains the equation

$P = 4897Y - 9\,657\,847$, where P is the median house price in pounds and Y is the calendar year, for example 2015.

(b) Use Dr Procter's equation to predict the median house price in Barking and Dagenham in

- 2016
- 2017. **[2]**

Professor Jackson uses a simpler model by using the data from 2014 and 2015 only to form a straight-line model.

(c) Find the equation Professor Jackson uses in her model. **[2]**

(d) Use Professor Jackson's equation to predict the median house price in Barking and Dagenham in

- 2016
- 2017. **[2]**

Professor Jackson carries out some research online. She finds some information about median house prices in Barking and Dagenham, which is shown in Fig. 11.2.

2016	2017
£290 000	£300 000

Fig. 11.2

(e) Comment on how well

- Dr Procter's model fits the data,
- Professor Jackson's model fits the data. **[2]**

(f) Explain which, if any, of the models is likely to be more reliable for predicting median house prices in Barking and Dagenham in 2020. **[1]**